

PRODUCT BACKLOG

TBDA · GROUP 2 · CeDInt

Energy Waste & Anomaly Detection

Revision following feedbacks

1. Product purpose

The delivered product is an interactive decision-support platform for CeDIn. It analyses historical electricity consumption, predicts normal consumption for supported zones/load types, compares actual and expected use, detects potentially wasteful or abnormal periods, and communicates energy, financial and CO2 impact.

2. Operational context and product boundaries

- Intended users: Building Manager, Energy/Sustainability Officer, Maintenance Technician and Sponsor/Technical Owner.
- The solution is a historical-data decision-support tool.
- It does not automatically control lighting, HVAC or other equipment.
- Real-time streaming, live-pipeline modification and specific hardware integration are outside the delivered product scope.
- The handover documentation must state assumptions, limitations, dependencies and future deployment requirements.

3. Product backlog principle

The backlog describes WHAT the client/user receives. Coding, data cleaning, resampling, model training, database configuration and chart programming are Sprint Backlog tasks derived from these product requirements, not product features by themselves.

4. Energy load scope

The product must explicitly identify the energy loads it analyses. The final circuit-to-load mapping must be validated and recorded in the data dictionary.

Load type	Scope	Product use	Requirement
Lighting	Included	Forecast + anomaly/waste	Separately identifiable where circuits exist
Emergency lighting	Included where present	Circuit-level analysis	Kept distinct from normal lighting
Plug/socket loads	Included	Forecast + anomaly/waste	Mapped by circuit/zone
HVAC electrical loads	Included where monitored	Forecast + anomaly/waste	No unavailable load is inferred
Hand dryers / other monitored loads	Included where validated	Anomaly/waste, forecasting if suitable	Each circuit has a documented type
Unknown/unmapped loads	Not silently aggregated	Scope/data limitation	Mapping must be validated first

5. UI/UX and non-functional requirements

- UI is designed around stakeholder tasks and personas.
- A mock-up/wireframe exists before implementation and receives stakeholder feedback.
- Dashboard navigation, filters, terminology and status indicators are consistent.
- Actual, expected and prediction-band values are visually distinguishable.
- Alert severity is understandable without knowledge of the ML model.
- Data-quality issues are distinguishable from energy-waste alerts.
- Outputs remain traceable to zone, circuit/load type and time period.
- Analysis is reproducible from documented inputs and configuration.
- Performance targets are agreed for the deployment context; the proposed overview target is under 3 seconds.
- Limitations, dependencies and future deployment responsibilities are documented.

6. Product Backlog

Current state: Sprint 2 is In Progress, Sprint 1 is Complete, Sprint 3–5 are Not Started. Priority uses Must / Should / Could.

ID	EPIC	AS A ...	I WANT TO ...	SO THAT ...	ACCEPTANCE CRITERIA	PRIORITY	SPRINT	STATUS
PB-01	Operational context	Building Manager	access the product in a defined historical-data analysis environment	understand how and when the solution is intended to be used	Purpose, users, analytical period and deployment context are stated	Must	S1	Complete
PB-02	Operational context	Sponsor	understand that the solution is a decision-support tool	avoid treating it as an automatic building-control system	No automatic control of lighting, HVAC or other equipment is included	Must	S1	Complete
PB-03	Operational context	Technical Owner	know the intended user roles	assign responsibility for interpretation and operation	Building Manager, Energy/Sustainability Officer, Maintenance Technician and Sponsor/Technical Owner are defined	Must	S1	Complete
PB-04	Operational context	Technical Owner	know the deployment and handover context	prepare an appropriate organisational deployment	Dependencies, assumptions and responsibilities are documented	Must	S2	In Progress
PB-05	Operational context	Sponsor	know which capabilities are excluded	avoid assuming real-time streaming, live pipeline changes or hardware integration are delivered	Explicit exclusions are listed	Must	S1	Complete
PB-06	Energy scope	Energy Analyst	see the energy load types covered by the product	know exactly which uses can be analysed and modelled	Load categories and circuit mapping are documented	Must	S2	In Progress
PB-07	Energy scope	Energy Analyst	analyse lighting consumption by zone and time	identify abnormal or unnecessary lighting use	Lighting circuits are identifiable and selectable	Must	S3	Not Started
PB-08	Energy scope	Energy Analyst	analyse emergency-lighting consumption	identify abnormal emergency-lighting behaviour	Emergency-lighting circuits remain distinguishable where present	Should	S3	Not Started
PB-09	Energy scope	Energy Analyst	analyse plug/socket loads	identify unnecessary equipment operation	Plug/socket circuits are identifiable by zone/circuit	Must	S3	Not Started
PB-10	Energy scope	Energy Analyst	analyse HVAC electrical consumption where monitored	identify abnormal heating/cooling-related electrical behaviour	Only actually monitored HVAC loads are included; unavailable loads are not inferred	Must	S3	Not Started

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PB-11	Energy scope	Energy Analyst	analyse hand dryers and other validated monitored loads	avoid hiding relevant loads inside an unexplained aggregate	Each additional circuit has a documented load type	Should	S3	Not Started
PB-12	Energy scope	Sponsor	see which load types are included and excluded	understand the limits of the delivered results	Included/excluded categories appear in the scope and data dictionary	Must	S2	In Progress
PB-13	Data foundation	Energy Analyst	access a consistent analytical dataset	use one common data foundation for modelling	Clean data are traceable to source data and documented transformations	Must	S2	In Progress
PB-14	Data foundation	Energy Analyst	understand the meaning, unit and scope of each relevant measurement	interpret model inputs and outputs correctly	Data dictionary defines relevant variables, units, zones and circuits	Must	S2	In Progress
PB-15	Data foundation	Energy Analyst	analyse data using the agreed time representation	avoid mixing incompatible temporal resolutions	Treatment of 2020–23 hourly and 2025 minute-level data is documented	Must	S2	In Progress
PB-16	Data foundation	Energy Analyst	use timestamps in the agreed local time basis	correctly interpret night, weekend and holiday behaviour	Europe/Madrid and daylight-saving treatment are documented and checked	Must	S2	In Progress
PB-17	Data foundation	Energy Analyst	identify missing, corrected and flagged observations	distinguish data limitations from real energy behaviour	Data-quality status remains traceable; abnormal values are not silently deleted	Must	S2	In Progress
PB-18	Data foundation	Sponsor	review the main data limitations	judge whether the dataset is suitable for analysis	A data-readiness summary lists gaps, limitations and assumptions	Must	S2	In Progress
PB-19	Context	Building Manager	see consumption during nights, weekends and holidays	identify energy use when the building is less occupied	Out-of-hours consumption is available by zone and relevant circuit	Must	S2	In Progress
PB-20	Context	Energy Analyst	understand standby/baseline consumption	distinguish necessary background use from potential waste	A documented baseline exists for relevant circuits	Must	S2	In Progress
PB-21	Context	Energy Analyst	understand weather and calendar relationships with consumption	interpret normal consumption before forecasting	Temperature and calendar effects are analysed	Should	S2	In Progress
PB-22	Context	Energy Analyst	know whether structural changes affect the modelling period	avoid treating outdated behaviour as current normal behaviour	Relevant structural breaks and training window are documented	Should	S3	Not Started
PB-23	Forecast	Energy Analyst	understand which variables are used to predict normal consumption	review the modelling logic and avoid information leakage	Target and features are documented; future information is excluded	Must	S3	Not Started
PB-24	Forecast	Building Manager	receive a forecast of normal consumption for each supported zone	compare actual use with expected use	Forecast is produced at the agreed product time resolution	Must	S3	Not Started
PB-25	Forecast	Building Manager	see normal consumption for relevant load types or circuits	identify which load explains an abnormal period	Outputs preserve the agreed zone/circuit/load-type level	Must	S3	Not Started
PB-26	Forecast	Energy Analyst	compare the ML approach with simple baselines	verify that model complexity adds measurable value	At least two baselines are evaluated on the same test period	Must	S3	Not Started
PB-27	Forecast	Sponsor	compare candidate models on unseen periods	make an evidence-based model-selection decision	Candidate models use the same time-based evaluation principle	Must	S3	Not Started
PB-28	Forecast	Energy Analyst	compare the ML approach	justify the final model choice	Same split and comparable metrics are used	Should	S3	Not Started

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			with a time-series alternative such as Prophet					
PB-29	Forecast	Sponsor	review forecast accuracy using agreed metrics	judge whether expected values are reliable for anomaly detection	MAE, MAPE and CV(RMSE), or final agreed metrics, are reported on unseen periods	Must	S3	Not Started
PB-30	Forecast	Building Manager	see an upper and lower expected-consumption band	distinguish normal variation from abnormal consumption	A documented prediction interval is produced and coverage is checked	Must	S3	Not Started
PB-31	Forecast	Modelling Team	reproduce model experiments	trace how the final model was selected	Runs, parameters and metrics are stored in an experiment log	Should	S3	Not Started
PB-32	Deep learning	Sponsor	see whether deep learning adds value over the selected ML baseline	make a justified decision on its use	A deep-learning candidate is evaluated with the same temporal test principle if required	Should	S3	Not Started
PB-33	Anomaly detection	Building Manager	receive an alert when actual consumption exceeds the expected range	identify potential abnormal consumption or waste	Alert includes zone/circuit and affected period	Must	S4	Not Started
PB-34	Anomaly detection	Building Manager	receive alerts for abnormal out-of-hours consumption	identify unexpected operation of lighting, plugs, HVAC or other loads	Detection uses documented baseline and calendar/occupancy rule	Must	S4	Not Started
PB-35	Anomaly detection	Maintenance Technician	distinguish data issues from waste alerts	avoid investigating sensor problems as real energy events	Data issues are shown separately with their reason	Must	S4	Not Started
PB-36	Anomaly detection	Building Manager	see the severity of each anomaly	prioritise events	Severity levels and thresholds are documented	Must	S4	Not Started
PB-37	Anomaly detection	Sustainability Officer	see excess energy, financial and CO2 impact	rank problems by environmental and financial importance	Relevant alerts show excess kWh, € and CO2	Must	S4	Not Started
PB-38	Anomaly detection	Sponsor	see detection performance on known events	assess whether the detector finds meaningful cases	Known cases are tested and false-alarm behaviour is reported	Should	S4	Not Started
PB-39	Anomaly detection	Building Manager	receive a manageable number of alerts	avoid alert fatigue	Threshold and trade-off between missed events and false alarms are documented	Should	S4	Not Started
PB-40	Dashboard	Building Manager	open a dashboard with main energy KPIs and alerts	understand the situation quickly	Overview shows agreed kWh, €, CO2, waste and alert indicators	Must	S4	Not Started
PB-41	Dashboard	Building Manager	filter by zone, load type/circuit and time period	focus on a specific problem	Filters update relevant views consistently	Must	S4	Not Started
PB-42	Dashboard	Building Manager	compare actual and expected consumption on one chart	see when use leaves the normal range	Actual, forecast, band and alert markers are shown together	Must	S4	Not Started
PB-43	Dashboard	Building Manager	see Storey 1 and its NE/NW/SE/SW zones	locate priority areas spatially	Four zones are identifiable and selectable	Should	S4	Not Started
PB-44	Dashboard	Maintenance Technician	view a sortable and filterable alert list	know what to inspect and in which order	List includes time, zone, circuit/load type, severity, € and CO2	Must	S4	Not Started
PB-45	Dashboard	Maintenance Technician	open an alert and see its surrounding context	understand the event before investigating	Alert detail shows context and a plain-language reason	Should	S4	Not Started
PB-46	Dashboard	Sustainability Officer	see recurring waste patterns by day and hour	identify systematic unnecessary consumption	Heatmap supports relevant zone/floor views and marks out-of-hours periods	Should	S4	Not Started
PB-47	Dashboard	Sustainability	see monthly savings	prioritise actions and communicate	Monthly € and CO2 totals and leading sources	Must	S4	Not Started

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		Officer	opportunities and main waste sources	impact	are displayed			
PB-48	Dashboard	Building Manager	see clear normal, warning and high-severity states	interpret results without understanding the ML model	Visual status rules and terminology are consistent	Must	S4	Not Started
PB-49	Dashboard	Building Manager	use the dashboard without inspecting datasets or code	make routine decisions from the delivered interface	Each main screen contains the information needed for its intended task	Must	S4	Not Started
PB-50	Dashboard	Sponsor	review UI mock-ups before implementation	validate structure and information hierarchy	All intended screens are mocked up and feedback is recorded	Must	S2	In Progress
PB-51	Dashboard	Product Team	test the dashboard with people outside the team	identify usability problems before delivery	Task-based usability test is performed and findings addressed or recorded	Should	S5	Not Started
PB-52	Dashboard	Sponsor	use historical demonstration mode	review the complete product without a live feed	A selected historical period can be replayed or demonstrated	Should	S5	Not Started
PB-53	Reporting	Sustainability Officer	export alert and savings information	share findings with management	Relevant filtered alerts and monthly impact can be exported	Could	S5	Not Started
PB-54	Handover	Technical Owner	access technical documentation	run, review and maintain the solution	Documentation covers setup, data, model, detection rules and dashboard	Must	S4–S5	Not Started
PB-55	Handover	Building Manager	access a concise user guide	use the product without detailed technical support	Guide includes screenshots and alert interpretation	Should	S5	Not Started
PB-56	Handover	Sponsor	receive a deployment and handover plan	understand requirements for future organisational operation	Plan states dependencies, hosting/data assumptions, retraining/monitoring, costs and responsibilities	Must	S5	Not Started
PB-57	Handover	Sponsor	receive reproducible analysis, code, documentation and data dictionary	review and reuse the methodology and results	Agreed deliverables are organised and reproducible from documented inputs	Must	S5	Not Started
PB-58	Handover	Sponsor	receive a final presentation and working demonstration	assess value, limitations and operational implications	Presentation covers problem, method, results, impact, deployment context and lessons learned	Must	S5	Not Started

7. Proposed UI structure

Screen	Main user	Required content	Main actions
Overview	Building Manager / Sponsor	Period; kWh; €; CO2; waste; open alerts	Select period; navigate
Actual vs Expected	Building Manager / Analyst	Actual; forecast; band; alerts	Filter zone/load/date
Floor / Zone	Building Manager	Storey 1; NE/NW/SE/SW; status	Select zone
Alert list	Maintenance	Time; zone; circuit/load; severity; €; CO2	Filter, sort, open
Alert detail	Maintenance	Context window; reason; affected load; impact	Inspect event
Waste heatmap	Sustainability	Day × hour waste; zone/floor	Find recurring patterns
Savings report	Sustainability	Monthly € / CO2; main sources	Review / report
Demo mode	Sponsor	Historical replay	Select period

8. Delivery and handover package

- Working dashboard covering the agreed scope.
- Final forecasting and anomaly-detection methodology with model-selection evidence.
- Clean analytical dataset or reproducible process for rebuilding it from agreed raw inputs.
- Data dictionary covering zones, circuits, load types, variables and units.
- Technical documentation covering setup, data, modelling, detection and dashboard.
- User guide with screenshots and alert interpretation.
- Deployment/handover plan covering dependencies, future live-data assumptions, hosting, retraining/monitoring, costs and responsibilities.
- Final presentation and working demonstration.
- Backup demonstration material where required.